

Reviewer Pack — Tensor Rank of Clause Incidence Matrix Bounds ACC⁰ Circuit ...

Entry 86fd76e5eb2d · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Tensor Rank of Clause Incidence Matrix Bounds ACC⁰ Circuit Size

Entry ID: 86fd76e5eb2d **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-19 15:38:57 UTC

1. Conjecture

Field A (mathematical branch): Tensor Rank over Finite Fields **Field B** (complexity object): ACC⁰ Circuit Size

Statement:

For any CNF formula with n variables, the tensor rank of its incidence matrix over $\text{GF}(2)$ is $\Theta(\log n)$ if and only if the formula can be computed by an ACC⁰ circuit of size $O(n^k)$ for some k . Conversely, if the tensor rank exceeds $\Omega(n^\epsilon)$, then no ACC⁰ circuit of size $n^{o(1)}$ can compute the formula.

Rationale (proposer's reasoning):

Tensor rank captures the minimal number of rank-1 tensors needed to represent the incidence matrix, which reflects the algebraic complexity of the CNF's interaction structure. This invariant may expose inherent limitations in ACC⁰'s ability to compress combinatorial dependencies, aligning with Williams' framework for circuit lower bounds.

Taxonomy category: ACC_LB_via_WILLIAMS (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 97f9d6748972b352

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	SAFE

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.95	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.choice([-1, 1]) * random.randint(1, n) for _ in range(3)]
            if len(set(clause)) == 3:
                clauses.append(clause)
        return clauses

    def incidence_matrix(clauses, n):
        m = len(clauses)
        A = [[0] * n for _ in range(m)]
        for i, clause in enumerate(clauses):
            for var in clause:
                if var > 0:
                    A[i][var - 1] = 1
                else:
                    A[i][-var - 1] = 1
        return A

    def tensor_rank(A):
        m, n = len(A), len(A[0])
        rank = 0
        for i in range(n):
            if any(A[j][i] == 1 for j in range(m)):
                rank += 1
        return rank

    n = random.randint(5, 40)
    clauses = generate_3cnf(n)
    A = incidence_matrix(clauses, n)
    tensor_rk = tensor_rank(A)
```

```

# Check if the formula can be computed by an ACC^0 circuit of size O(n^k) for some k
acc0_circuit_size = n ** (random.random() * 2 + 1)

return {
    "metric_name": "tensor_rank",
    "metric_value": tensor_rk,
    "instances_tested": 1,
    "conjecture_holds": tensor_rk <= math.log(n, 2) and acc0_circuit_size >= n ** (math.log(n, 2) + 1),
    "counterexample": "" if tensor_rk <= math.log(n, 2) else f"tensor_rank={tensor_rk} > log2(n)"
}

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [random.randint(1000, 9999) for _ in range(30)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["counterexample"] == "" for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif any(r["counterexample"] != "" for r in results):
        first_failing_seed = next(r["seed"] for r in results if r["counterexample"] != "")
        print(f"RESULT: FALSIFIED counterexample={r['counterexample']} first_failing_seed={first_failing_seed}")
    else:
        print("RESULT: INCONCLUSIVE")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

also, 'counterexample': 'tensor_rank=15 > log2(15)'}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 7, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 25, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 23, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 9, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 38, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 16, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 36, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 19, 'instances_tested': 1, 'conjecture_holds': True}
TRIAL: {'metric_name': 'tensor_rank', 'metric_value': 17, 'instances_tested': 1, 'conjecture_holds': True}

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2b97f68.py", line 81, in <module>
    first_failing_seed = next(r["seed"] for r in results if r["counterexample"] != "")
                        ~~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_e2b97f68.py", line 81, in <genexpr>

```

```
first_failing_seed = next(r["seed"] for r in results if r["counterexample"] != "")
                        ~^^^^^^^^^
```

KeyError: 'seed'

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed before producing valid data, rendering all trial results unreliable. | next:
Re-run test with increased time limits and deterministic seed configuration

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	112770
2	propose	ollama_remote	qwen3:8b	0	0	68008
3	preregistration	ollama_remote	qwen3:8b	0	0	30381
4	novelty	ollama_remote	qwen3:8b	0	0	24054
5	novelty	ollama_remote	qwen3:8b	0	0	17977
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10335
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10922
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10798
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8616
10	judge	ollama_remote	qwen3:8b	0	0	19334

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 313195 ms total latency. Provider mix:
{'ollama_remote': 10}

(full prompt+response transcripts available in *research/audit/86fd76e5eb2d.jsonl*)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in *research/audit/86fd76e5eb2d.jsonl* for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: *research/pvsnp_sandbox/test_*.py* (per-cycle)

Replay tarball: *research/replay/86fd76e5eb2d.tar.gz* (if generated)